

Sheridan College
School of Applied Science and Engineering

Class Plan: Fall 2026

Course: MATH15325D - Linear Algebra

Professor: Cyrus Hosseini PhD PEng

This document presents an initial class plan for this course. The dates and activities indicated below are subject to change or modification due to unforeseen circumstances such as snowstorms, absences, governmental or college closures/restrictions. Any changes to this plan will be announced **in class or via SLATE**. Please monitor the announcements page throughout the semester.

1 EVALUATION PLAN

The final grade of this course will be comprised of the following elements:

No.	Evaluation Component	Scheme A
1	4 Quizzes (5 % each)	20 %
2	3 Assignments (4 %, 3%, 3%)	10 %
4	Test 1	20 %
5	Test 2	20 %
6	Final Exam	30 %

Note: The passing grade in this course is 60%. Accreditation Units: Mathematics 100%.

2 MATLAB ACCESS AND USE:

MATLAB is extensively used throughout out the course. Students are given access to MATLAB by the College and are required to be ready to access and use it anytime on their Laptops during classrooms or tutorials. Students' mastery of MATLAB has significant impact on their future courses where MATLAB will be widely utilized.

Submission of MATLAB Assignments

1. All MATLAB Assignments must be submitted online to the respective Drop Box. MATLAB Assignments must be submitted before the **due date/time** the latest to be considered for full mark. **A single PDF file must contain all your theoretical analysis, MATLAB code, and MATLAB Simulation Output.**
2. A late submission will attract a 10% deduction per day, to a maximum of 30%. Any assignment submitted after 3 days late will be awarded a grade of 0.

3 COURSE MATERIALS AND OUTLINE

The textbook used for the course is:

Elementary Linear Algebra (Applications Version), Anton, Howard; Rorres, Chris and Kaul, Anton, WILEY, 12th edition

4 LEARNING OUTCOMES

To achieve the critical performance, students will have demonstrated the ability to:

1. Perform algebraic operations on complex variables in rectangular, polar, trigonometric and geometric forms (KB.1).
2. Solve determined linear systems of equations by elementary row operations, matrix operations, Gaussian elimination, inverse matrices and diagonalization (KB.1).
3. Solve undetermined linear systems of equations by determining rank, nullity and fundamental matrix spaces (KB.1).
4. Perform matrix operations on matrices including matrix algebra, inverse matrices, determinants using cofactor expansion, row reduction and Kramer's rule (KB.1).
5. Apply concepts from Euclidean vector spaces such as basis, change of basis, norms, dot and cross products and orthogonality to analyze the geometry of linear systems in both geometrical and component forms (KB.1).
6. Apply concepts such as subspaces, solution spaces, linear independence, choice of basis and change of basis to analyze general real and complex vector spaces (KB.1).
7. Evaluate eigenvalues and eigenvectors of real and complex $n \times n$ matrices (KB.1).
8. Diagonalize real and complex $n \times n$ matrices (KB.1).
9. Apply concepts from inner product spaces to discuss norm of a vector, unit vectors, angles, orthogonal projections and orthogonal matrices and find orthogonal bases using Gram-Schmidt process (KB.1).
10. Perform algebraic operations on matrices and arrays using numerical analysis software (ET.1, ET.3).

5 ACADEMIC PROCEDURES AND GUIDELINES

The students are expected to abide by the Student Code of Conduct, Academic Integrity, Harassment and Discrimination and any other applicable policies put in place by Sheridan College. In addition, the following should be considered:

1. Students should attend class on time and be prepared to work for the entire scheduled time. Regular attendance is necessary to participate in classroom activities, and to receive evaluation, feedback, and instructions.
2. Assignments are due as advised by the instructor. Late submission will attract a 10% deduction per day, to a maximum of 30%. Assignments submitted after 3 days late will be awarded a grade of "0".

3. Please refer to the supporting documents posted in the Student Success in Engineering Virtual Community on SLATE. Students who belong to other schools (ie: Applied Computing, Chemical, Business, etc.) will need to follow their Faculty Office's missed evaluation process.
4. *Missed evaluations will result in a grade of zero.* See the process document and Verification of Injury or Illness Form posted on the Student Success In Engineering Virtual Community page on SLATE.
5. It is the student's responsibility to obtain course material and notes for any missed classes. Being absent is no excuse for not knowing.
6. Students are expected to use course texts and reference texts for independent study. The content of any evaluation component (assignments, quizzes, tests, etc.) may include references from these sources.
7. Students are responsible for monitoring own progress in the course. Grades will be posted on SLATE.
8. Only some "acceptable" types of calculators are allowed during tests. Other electronic devices (including cellphones) are not permitted. Sharing of calculators during evaluations is not allowed.
9. Work submitted for tests, assignments and laboratory reports should be the individual students. In some cases, the instructor may ask students to work in groups and submit group work. Plagiarized work will be punishable according to college and faculty guidelines.
10. Policies regarding appeals and dishonesty are outlined in the student handbook. It is the student's responsibility to be aware of and comply with these policies. Some information is also provided on SLATE.
11. Students are expected to conduct themselves in way that displays full respect for everybody in the room to ensure that a positive learning environment is maintained.
12. Students are prohibited to use communication devices and MISUSE laptops during class sessions.
13. Communication between the professor and students will be conducted through SLATE.
14. Final Exam, Test and Quizzes will be closed book and formulas will be provided at the discretion of professor.
15. Students are encouraged to make use of consultation time.

Week #	Week starting on	Lecture Topics	Classwork / activities
1	7-Sep-2026	Algebra of complex numbers in rectangular, polar and exponential forms	Course introduction – class plan discussion
2	14-Sep-2026	Systems of linear equations, matrices and matrix operations, Gauss elimination	
3	21-Sep-2026	Matrix algebra, Special matrices	Assignment 1 (4%)
4	28-Sep-2026 No classes on Sep. 30 (National Truth and Reconciliation Day)	Properties of determinants, determinants by row reduction, cofactor expansion and Cramer's rule	Assignment 1 is due. Quiz 1 - (5%)
5	5-Oct-2026 No classes on Oct. 12 (Thanksgiving)	Euclidean vector spaces, norm, distance, dot product	Quiz 2 - (5%)
6	12-Oct-2026	Orthogonality, Geometry of lines and planes Cross Product, General vector spaces, Subspaces	
7	19-Oct-2026	No Classes on Week 7 – Test 1 on October 19, 2026 - (20 %) – Material from Week 1-6	
-	26-Oct-2026	Reading Week – No classes planned	
8	2-Nov-2026	Linear Independence, Dimension, Basis and Change of Basis	Assignment 2 (3%)
9	9-Nov-2026	Row, column, and null spaces	Assignment 2 is due. Quiz 3 - (5%)
10	16-Nov-2026	Rank, Nullity, and Fundamental Matrix Spaces and Solution of Undetermined Systems	Test 2 - (20 %) – Material from Week 5-10
11	23-Nov-2026	Eigenvalues and Eigenvectors, diagonalization	Assignment 3 (3%)
12	30-Nov-2026	Complex vector spaces, Inner product spaces, norm, and distance	Assignment 3 is due. Quiz 4 - (5%)
13	7-Dec-2026	Angles and Orthogonality, Gram-Schmidt Process	
14	14-Dec-2026	No Classes on Week 14 – Final Exam on December 14, 2026 - (30%) - Comprehensive Exam, All materials from Week 1 -13	